

How to Learn to Encode Expertise (Without an Expert in the Room)

Goal: learn to take a domain that already has practitioners — investment ESG assessment — and capture its reasoning well enough to build a tool, when you cannot sit with a sustainability or investment person.

The premise to flip: you are not learning the expertise from scratch. You are doing **archaeology on expertise that is already written down**. Experts leave their judgement in artefacts. Your job is to reverse-engineer those artefacts.

1. Read the reasoning, not the rules

Rules tell you *what*. The reasoning tells you *why*, which is the only thing that generalises into a design.

- **"Basis for conclusions" documents.** ISSB, GRI and EFRAG all publish why they chose what they chose, and what they rejected. This is the closest thing to having the expert in the room. Most people skip these — they are the highest-value reading there is.
- **Consultation responses.** When a standard goes out for comment, practitioners write in saying "this will not work because...". That is a free map of real-world edge cases and disagreements.
- **Regulator Q&A and enforcement notices.** These show where firms actually got it wrong, which is where your model needs to be sharpest.

2. Read practitioner artefacts, not textbooks

Textbooks teach the domain. Artefacts show the *encoding* — which is what you need.

- Public fund ESG policies and responsible-investment reports: how a real investor states scope, exclusions, and escalation.
- ILPA DDQ, PRI reporting framework, LP questionnaires: these are literally someone's encoded assessment model, already written down.
- Published rating methodologies (Sustainalytics, MSCI, S&P CSA): read the *methodology* documents, not the scores. They make weighting and normalisation choices explicit.
- Actual DDQ responses and ESG due-diligence reports from consultancies: hard to get for free, but templates circulate widely.

3. Build a disagreement map

Consensus is easy to find and low-value. **The encoding value is where experts disagree** — whether biodiversity is material to retail, whether Scope 3 counts, whether exposure should

weight linearly.

Find the ten live disagreements in your domain and decide each one explicitly. That set of decisions *is* your framework. Sources: consultation responses, conference panels, public debates between practitioners.

4. Use AI as interrogator, not oracle

- Have it play a sceptical CIO and a defensive sustainability officer, then attack your scoring model.
- Ask it to argue the *opposite* of each design choice, and to state what evidence would change its mind.
- Never ask it "what is the right weighting" — it will invent one confidently. Ask it "what are the arguments for and against each weighting option" instead.

5. Backtest against reality (the strongest substitute for an interview)

Reality is the expert you cannot interview.

- Take companies with **known ESG failures** — governance scandals, safety disasters, greenwashing enforcement. Run them through your model using only pre-event information. If your framework would not have flagged them, the model is wrong.
- Take well-regarded companies and check you do not score them Severe. False positives are as damaging as misses, because they destroy trust in the output.

This is how you calibrate without a human in the loop.

6. Structured learning, if you want the floor

- **CFA Certificate in ESG Investing** — the standard practitioner credential.
- **GRI-certified training** — strongest on impact materiality.
- **SASB Fundamentals (FSA) credential** — strongest on financial materiality, which is the investor lens.

These compress the domain into a syllabus and give you the vocabulary to read everything above properly. They do not substitute for the artefacts.

The honest summary

You can get to roughly 80% without an expert, through artefacts, disagreement mapping, and backtesting. The remaining 20% is tacit: knowing which edge case matters to *this* fund's mandate, and how much weight a real investment committee would place on it.

That part comes from a real deal, not from reading.

The fastest shortcut is still one hour with someone who does this daily. A warm intro to any sustainability analyst or ESG-focused portfolio manager beats three months of solo reading.

If you can get that hour, take it — and bring your disagreement map, not your questions.

Checklist

- Read one "basis for conclusions" document end to end.
- Read one fund's published ESG policy and RI report.
- Read one rating agency methodology document (not the scores).
- List ten live disagreements in the domain and decide each.
- Backtest the model against five known ESG failures.
- Backtest against five well-regarded companies to check for false positives.
- Red-team the scoring assumptions with AI as a hostile reviewer.